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his document contains diploma engineering maths-1 syllabus. help me make study material for each unit. Act as a diploma engineering teacher and create study material for diploma engineering student who are beginners s

Unit 2: Trigonometry — Complete Study Material

Welcome to Unit 2! As your engineering mathematics teacher, I have structured this material step-by-step specifically for beginners. Trigonometry is a fundamental tool across civil, mechanical, electrical, and computer engineering. PDF + 1

2.1 Units of Angles (Degree and Radian)

Concepts & Definitions

An **angle** is formed when a ray rotates about its initial position.

Degree System ($^\circ$): A full circular rotation is divided into 360 equal parts called degrees (360°).

Radian System (rad or superscript ^{c}): A radian is the angle subtended at the center of a circle by an arc whose length is equal to the radius of the circle. A full rotation equals $2\pi \text{ radians}$.

Conversion Formulae

$$\pi \text{ radians} = 180^\circ$$

1. **Degree to Radian:** Multiply by $\frac{\pi}{180}$

$$\theta \text{ (in radians)} = \theta^\circ \times \frac{\pi}{180}$$

2. **Radian to Degree:** Multiply by $\frac{180}{\pi}$

$$\theta \text{ (in degrees)} = \theta \text{ (rad)} \times \frac{180}{\pi}$$

Step-by-Step Solved Examples

Example 1: Convert 45° into radians.

Solution:

$$\theta = 45 \times \frac{\pi}{180} = \frac{45\pi}{180} = \frac{\pi}{4} \text{ radians}$$

Example 2: Convert $\frac{2\pi}{3}$ radians into degrees.

Solution:

$$\theta = \left(\frac{2\pi}{3}\right) \times \frac{180}{\pi} = \frac{2 \times 180}{3} = 120^\circ$$

2.2 Trigonometric Functions

Standard Ratios in a Right-Angled Triangle

Consider a right triangle with angle θ, Opposite side (O), Adjacent side (A), and Hypotenuse (H):

$$\sin\theta = \frac{O}{H}, \quad \cos\theta = \frac{A}{H}, \quad \tan\theta = \frac{O}{A}$$

$$\csc\theta = \frac{H}{O} = \frac{1}{\sin\theta}, \quad \sec\theta = \frac{H}{A} = \frac{1}{\cos\theta}, \quad \cot\theta = \frac{A}{O} = \frac{1}{\tan\theta}$$

Fundamental Trigonometric Identities

1. $\sin^2\theta + \cos^2\theta = 1$
2. $1 + \tan^2\theta = \sec^2\theta$
3. $1 + \cot^2\theta = \csc^2\theta$

Signs of Trigonometric Ratios in Quadrants (ASTC Rule)

First Quadrant (0° to 90°): All functions are positive ($+$).

Second Quadrant (90° to 180°): Sine and $\csc$ are positive ($+$).

Third Quadrant (180° to 270°): Tangent and $\cot$ are positive ($+$).

Fourth Quadrant (270° to 360°): Cosine and $\sec$ are positive ($+$).

*(Mnemonic: **A**ll **S**chool **T**eachers **C**razy or **A**dd **S**ugar **T**o **C**offee)*

2.3 Allied, Compound, and Multiple/Submultiple Angles

1. Allied Angles

Angles like $(90^\circ \pm \theta)$, $(180^\circ \pm \theta)$, $(270^\circ \pm \theta)$, $(360^\circ \pm \theta)$, and $(-\theta)$.

Rule for Angle $(180^\circ \pm \theta)$ and $(360^\circ \pm \theta)$: Trigonometric function remains unchanged. Determine the sign using ASTC.

$\sin(180^\circ - \theta) = \sin\theta$ (Q2: Sine is positive)

$\cos(180^\circ - \theta) = -\cos\theta$ (Q2: Cosine is negative)

$\sin(-\theta) = -\sin\theta$

$\cos(-\theta) = \cos\theta$

Rule for Angle $(90^\circ \pm \theta)$ and $(270^\circ \pm \theta)$: Function changes ($\sin \leftrightarrow \cos$, $\tan \leftrightarrow \cot, \sec \leftrightarrow \csc$). Determine the sign using ASTC.

$$\sin(90^\circ + \theta) = \cos\theta$$

$$\cos(90^\circ + \theta) = -\sin\theta$$

2. Compound Angle Formulae

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

3. Multiple Angles (2θ, 3θ)

Double Angle Formulae:

$$\sin(2\theta) = 2\sin\theta\cos\theta = \frac{2\tan\theta}{1 + \tan^2\theta}$$

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta = \frac{1 - \tan^2\theta}{1 + \tan^2\theta}$$

$$\tan(2\theta) = \frac{2\tan\theta}{1 - \tan^2\theta}$$

Useful Half-Angle Adaptations:

$$1 + \cos(2\theta) = 2\cos^2\theta \implies \cos^2\theta = \frac{1 + \cos(2\theta)}{2}$$

$$1 - \cos(2\theta) = 2\sin^2\theta \implies \sin^2\theta = \frac{1 - \cos(2\theta)}{2}$$

Step-by-Step Solved Examples

Example 3: Find the value of $\sin(105^\circ)$ using compound angle formulae.

Solution:

1. Express 105° as a sum of standard angles: $105^\circ = 60^\circ + 45^\circ$.
2. Apply $\sin(A+B) = \sin A \cos B + \cos A \sin B$:

$$\sin(105^\circ) = \sin(60^\circ + 45^\circ) = \sin 60^\circ \cos 45^\circ + \cos 60^\circ \sin 45^\circ$$

3. Substitute standard values ($\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 45^\circ = \frac{1}{\sqrt{2}}$, $\cos 60^\circ = \frac{1}{2}$, $\sin 45^\circ = \frac{1}{\sqrt{2}}$):

$$\sin(105^\circ) = \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{\sqrt{2}}\right) = \frac{\sqrt{3}+1}{2\sqrt{2}}$$

Example 4: Prove that $\frac{\sin(2\theta)}{1+\cos(2\theta)} = \tan\theta$.

Solution:

1. Start with Left Hand Side (LHS):

$$\text{LHS} = \frac{\sin(2\theta)}{1+\cos(2\theta)}$$

2. Substitute multiple angle formulae $\sin(2\theta) = 2\sin\theta\cos\theta$ and $1 + \cos(2\theta) = 2\cos^2\theta$:

$$\text{LHS} = \frac{2\sin\theta\cos\theta}{2\cos^2\theta}$$

3. Cancel common terms (2 and $\cos\theta$):

$$\text{LHS} = \frac{\sin\theta}{\cos\theta} = \tan\theta = \text{RHS}$$

4. Hence proved.

2.4 Graphs of Sine and Cosine & 2.5 Periodic Functions

Periodic Functions

A function $f(x)$ is called **periodic** if there exists a positive real number T such that $f(x+T) = f(x)$ for all x. The smallest such positive value T is called the **fundamental period**.

$\sin x$ and $\cos x$ have a period of 2π (360°).

$\tan x$ has a period of π (180°).

General Rule: Period of $\sin(kx)$ or $\cos(kx)$ is $\frac{2\pi}{|k|}$.

Graph Properties of $\sin x$ and $\cos x$

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Property	$y = \sin x$	$y = \cos x$
Domain	$(-\infty, \infty)$	$(-\infty, \infty)$
Range	$[-1, 1]$	$[-1, 1]$
Period	2π	2π
Value at 0	0	1
Symmetry	Odd Function ($\sin(-x) = -\sin x$)	Even Function ($\cos(-x) = \cos x$)

2.6 Sum and Factor Formulae

Factorization Formulae (Converting Sum/Difference into Product)

$$\sin C + \sin D = 2\sin\left(\tfrac{C+D}{2}\right)\cos\left(\tfrac{C-D}{2}\right)$$

$$\sin C - \sin D = 2\cos\left(\tfrac{C+D}{2}\right)\sin\left(\tfrac{C-D}{2}\right)$$

$$\cos C + \cos D = 2\cos\left(\tfrac{C+D}{2}\right)\cos\left(\tfrac{C-D}{2}\right)$$

$$\cos C - \cos D = -2\sin\left(\tfrac{C+D}{2}\right)\sin\left(\tfrac{C-D}{2}\right) = 2\sin\left(\tfrac{C+D}{2}\right)\sin\left(\tfrac{D-C}{2}\right)$$

Defactorization Formulae (Converting Product into Sum/Difference)

$$2\sin A\cos B = \sin(A+B) + \sin(A-B)$$

$$2\cos A\sin B = \sin(A+B) - \sin(A-B)$$

$$2\cos A\cos B = \cos(A+B) + \cos(A-B)$$

$$2\sin A\sin B = \cos(A-B) - \cos(A+B)$$

Step-by-Step Solved Example

Example 5: Prove that $\frac{\sin 5x+\sin 3x}{\cos 5x+\cos 3x} = \tan 4x.$

Solution:

1. Apply factor formula to numerator ($\sin C + \sin D$) with $C = 5x, D = 3x$:

$$\sin 5x + \sin 3x = 2\sin\left(\frac{5x+3x}{2}\right)\cos\left(\frac{5x-3x}{2}\right) = 2\sin(4x)\cos(x$$

2. Apply factor formula to denominator ($\cos C + \cos D$) with $C = 5x, D = 3x$:

$$\cos 5x + \cos 3x = 2\cos\left(\frac{5x+3x}{2}\right)\cos\left(\frac{5x-3x}{2}\right) = 2\cos(4x)\cos(x$$

3. Divide numerator by denominator:

$$\text{LHS} = \frac{2\sin(4x)\cos(x)}{2\cos(4x)\cos(x)} = \frac{\sin(4x)}{\cos(4x)} = \tan 4x = \text{RHS}$$

2.7 Inverse Trigonometric Functions

Concept

If $\sin\theta = x$, then $\theta = \sin^{-1} x$ (read as "sine inverse x").

> $\sin^{-1} x$ represents an **angle** whose sine is x.

Principal Value Branches

Function	Domain	Principal Value Range
$\sin^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$(-\infty, \infty)$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Important Inverse Properties

1. $\sin^{-1}(\sin\theta) = \theta, \quad \sin(\sin^{-1} x) = x$
2. $\sin^{-1}(-x) = -\sin^{-1} x$
3. $\cos^{-1}(-x) = \pi - \cos^{-1} x$
4. $\tan^{-1}(-x) = -\tan^{-1} x$
5. $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$
6. $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$
7. $\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right) \quad (\text{if } xy < 1)$

Step-by-Step Solved Example

Example 6: Evaluate $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$.

Solution:

1. $\tan^{-1}(1) = \frac{\pi}{4}$ (since $\tan\frac{\pi}{4} = 1$).
2. Using $\cos^{-1}(-x) = \pi - \cos^{-1} x$:

$$\cos^{-1}\left(-\frac{1}{2}\right) = \pi - \cos^{-1}\left(\frac{1}{2}\right) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

3. Using $\sin^{-1}(-x) = -\sin^{-1} x$:

$$\sin^{-1}\left(-\frac{1}{2}\right) = -\sin^{-1}\left(\frac{1}{2}\right) = -\frac{\pi}{6}$$

4. Combine all values:

$$\text{Total} = \frac{\pi}{4} + \frac{2\pi}{3} - \frac{\pi}{6}$$

5. Take LCM (12):

$$\text{Total} = \frac{3\pi + 8\pi - 2\pi}{12} = \frac{9\pi}{12} = \frac{3\pi}{4}$$

Unit 2 Practice Exercise

Try these problems to test your understanding before moving to Unit 3!

1. Convert 300° into radians.
2. Find the fundamental period of $f(x) = \cos(4x)$.
3. Prove that $\frac{\cos 2\theta}{1-\sin 2\theta} = \frac{1+\tan\theta}{1-\tan\theta}$.
4. Simplify $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right)$.

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Where would you like to go next?

Proceed to Unit 3: Vectors

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Get solution key for Unit 2 practice exercise

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